Plant Leaf Disease Classification: Model Performance Report

Dataset Overview

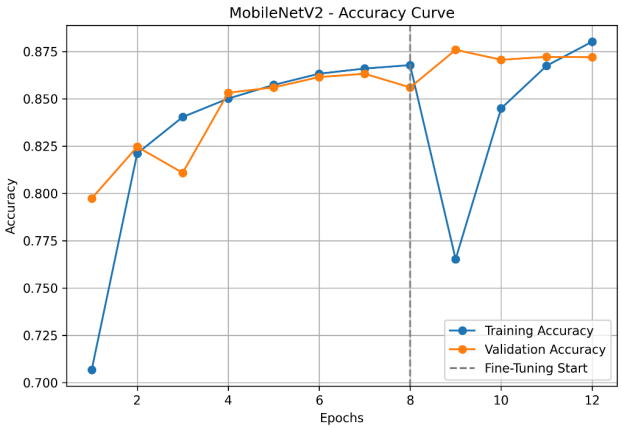
- **Classes**: 10 Tomato Disease Classes
- **Training Dataset**: 18,345 images
- **Validation Dataset**: 4,585 images
- **Training Strategy**: Two-stage transfer learning (Stage 1: Feature Extraction, Stage 2: Fine-Tuning top layers)

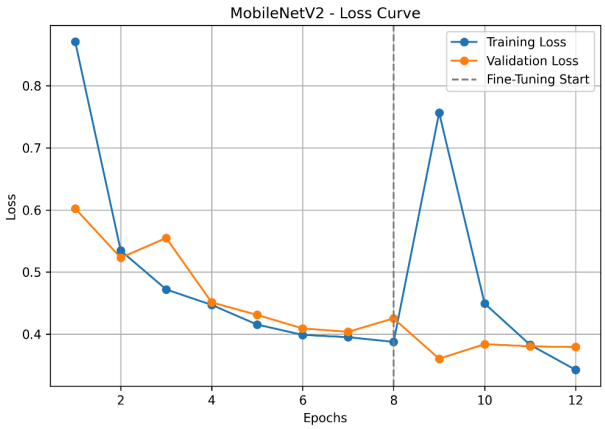
Model 1: MobileNetV2

Metric / Parameter	Value
Input Resolution	224 × 224 × 3
Total Parameters	2,269,962
Stage 1 Peak Validation Accuracy	86.32%
Stage 2 Peak Validation Accuracy	**87.59%**
Final Validation Loss	**0.3605**

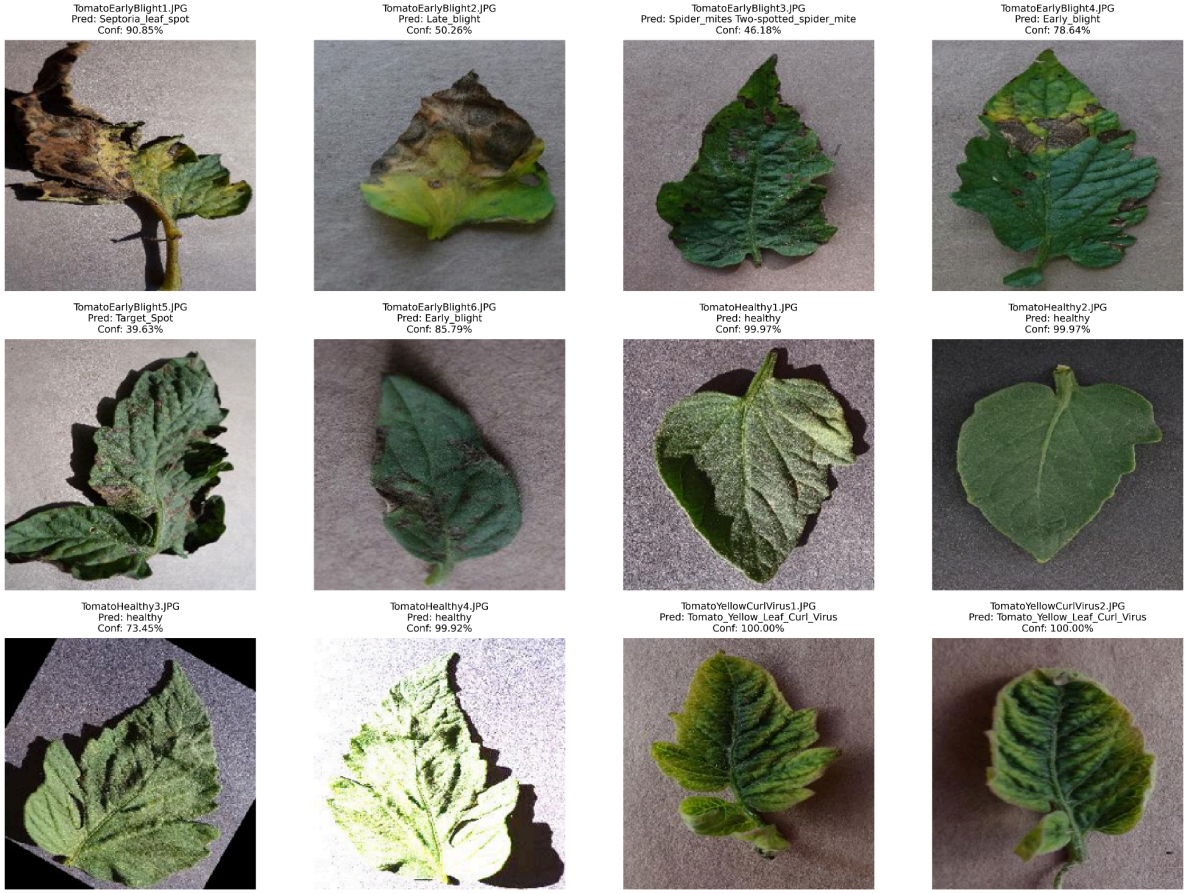
Summary

MobileNetV2 provided a lightweight baseline using inverted residual blocks and depthwise separable convolutions. During Stage 1 feature extraction, the validation accuracy peaked at 86.32% with a loss of 0.4039. Unfreezing top layers during Stage 2 fine-tuning offered a modest performance boost, reaching 87.59% validation accuracy and reducing validation loss to 0.3605 before early stopping triggered.





Test Set Predictions - MobileNetV2

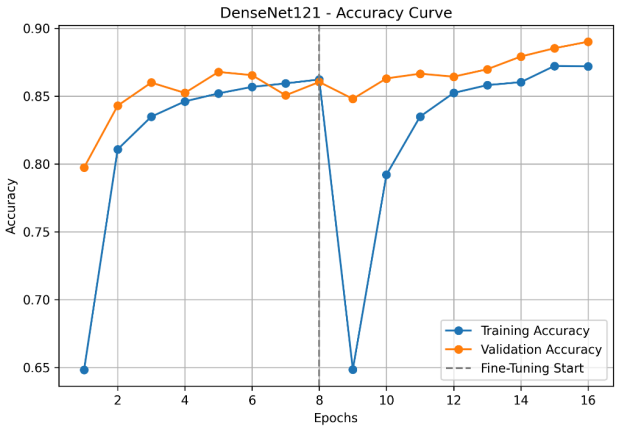


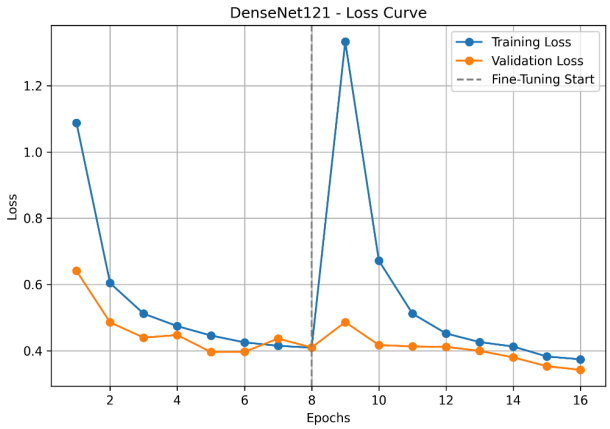
Model 2: DenseNet121

Metric / Parameter	Value
Input Resolution	224 × 224 × 3
Total Parameters	7,048,074
Stage 1 Peak Validation Accuracy	86.78%
Stage 2 Peak Validation Accuracy	**89.01%**
Final Validation Loss	**0.3424**

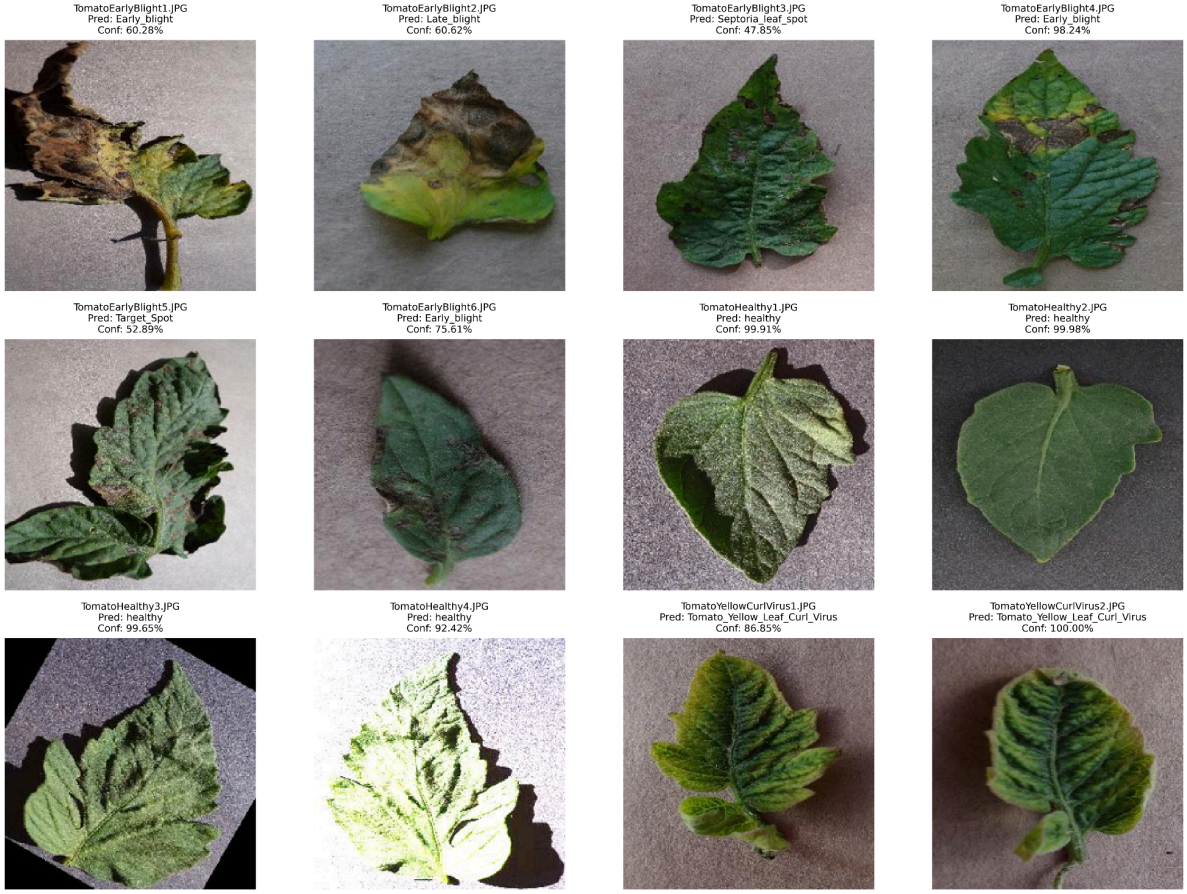
Summary

DenseNet121 connects each layer to every other layer in a feed-forward fashion, encouraging feature reuse across scale levels. Feature extraction topped out at 86.78% validation accuracy. Stage 2 fine-tuning demonstrated steady epoch-over-epoch progress, pushing validation accuracy to 89.01% and dropping validation loss to 0.3424 at Epoch 8.





Test Set Predictions - DenseNet121

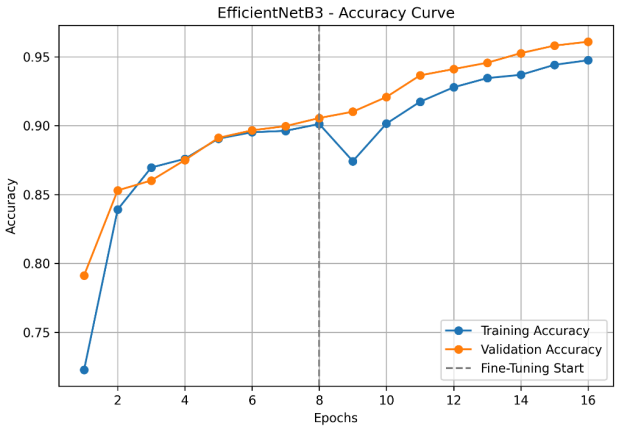


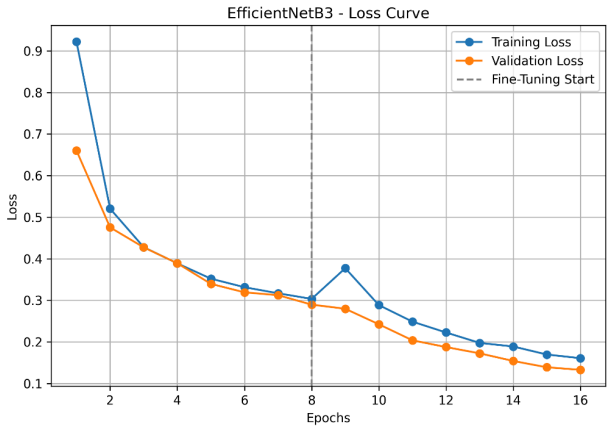
Model 3: EfficientNetB3

Metric / Parameter	Value
Input Resolution	300 × 300 × 3
Total Parameters	~12,238,000
Stage 1 Peak Validation Accuracy	90.56%
Stage 2 Peak Validation Accuracy	**96.10%**
Final Validation Loss	**0.1327**

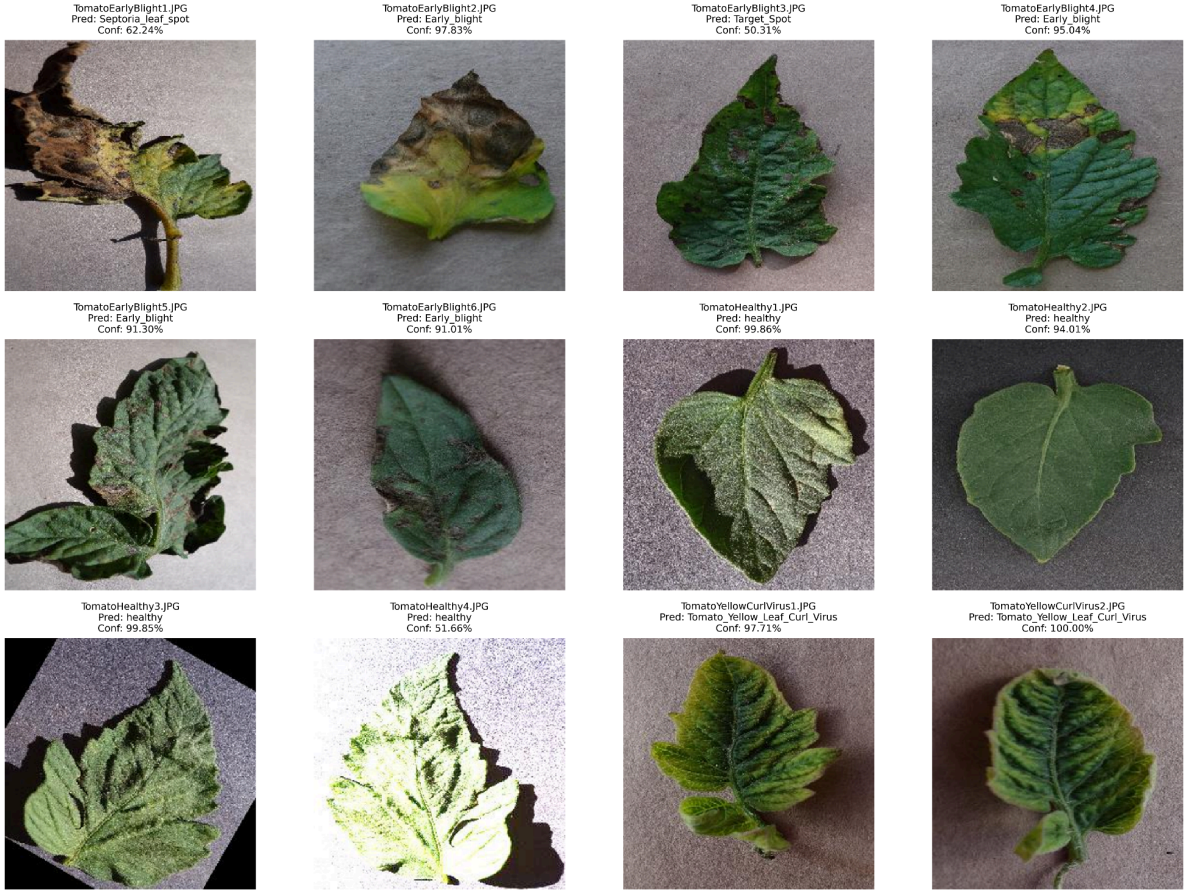
Summary

EfficientNetB3 utilizes compound scaling to systematically balance network depth, width, and input image resolution. Operating at 300×300 resolution, the model achieved 90.56% validation accuracy in Stage 1 alone. During Stage 2 fine-tuning, performance scaled cleanly across all 8 epochs, reaching a peak validation accuracy of 96.10% and a low validation loss of 0.1327.





Test Set Predictions - EfficientNetB3



Final Comparative Summary

Model	Resolution	Parameters	Stage 1 Peak Val Acc	Stage 2(Fine Tuning) Peak Val Acc	Best Val Loss
MobileNetV2	224 × 224	2.27 M	86.32%	87.59%	0.3605
DenseNet121	224 × 224	7.05 M	86.78%	89.01%	0.3424
EfficientNetB0 (Prior Baseline)*	224 × 224	5.33 M	90.10%	93.13%	0.1978
EfficientNetB3	300 × 300	12.24 M	**90.56%**	**96.10%**	**0.1327**